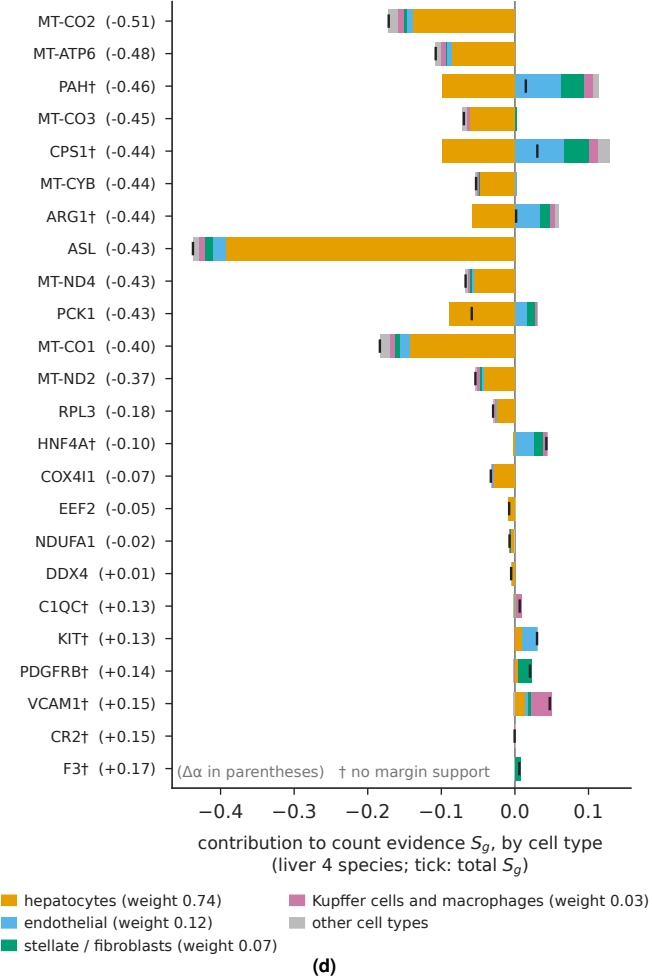
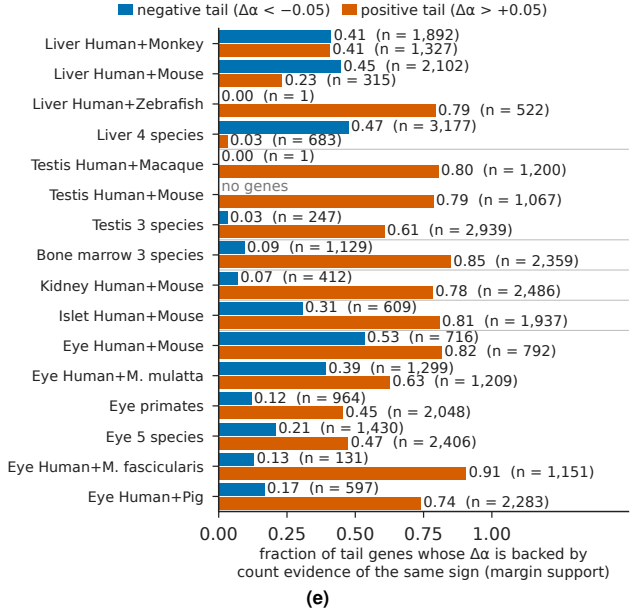
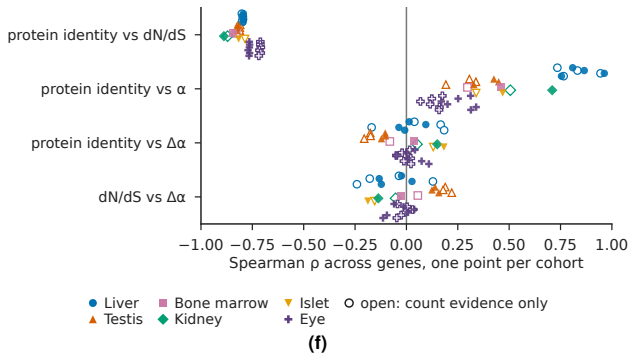

MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in parentheses) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count evidence S_g, by cell type
(liver 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types

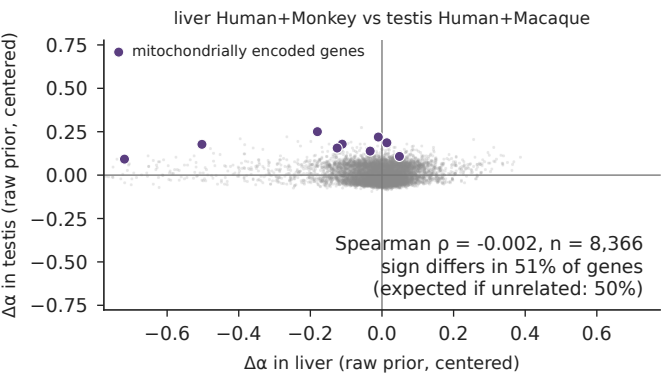
(d)


negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
Liver Human+Monkey 0.41 (n = 1,892) 0.41 (n = 1,327)
Liver Human+Mouse 0.45 (n = 2,102) 0.23 (n = 315)
Liver Human+Zebrafish 0.00 (n = 1) 0.79 (n = 522)
Liver 4 species 0.47 (n = 3,177) 0.03 (n = 683)
Testis Human+Macaque 0.00 (n = 1) 0.80 (n = 1,200)
Testis Human+Mouse no genes 0.79 (n = 1,067)
Testis 3 species 0.03 (n = 247) 0.61 (n = 2,939)
Bone marrow 3 species 0.09 (n = 1,129) 0.85 (n = 2,359)
Kidney Human+Mouse 0.07 (n = 412) 0.78 (n = 2,486)
Islet Human+Mouse 0.31 (n = 609) 0.81 (n = 1,937)
Eye Human+Mouse 0.53 (n = 716) 0.82 (n = 792)
Eye Human+M. mulatta 0.39 (n = 1,299) 0.63 (n = 1,209)
Eye primates 0.12 (n = 964) 0.45 (n = 2,048)
Eye 5 species 0.21 (n = 1,430) 0.47 (n = 2,406)
Eye Human+M. fascicularis 0.13 (n = 131) 0.91 (n = 1,151)
Eye Human+Pig 0.17 (n = 597) 0.74 (n = 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by
count evidence of the same sign (margin support)

(e)


protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort
Liver
Testis
Bone marrow
Kidney
Islet
Eye
open: count evidence only

(f)


liver Human+Monkey vs testis Human+Macaque
mitochondrially encoded genes
Δα in testis (raw prior, centered)
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)
liver Human+Mouse vs testis Human+Mouse
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)

(g)